FoodLogAthl-218: A Real-World Benchmark for Food Recognition from Dietary Logging Applications

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Abstract

Food recognition benchmarks are still dominated by web-collected or heavily curated imagery, which makes it difficult to assess how models behave on the casual, long-tailed, multi-dish, and temporally structured images encountered in dietary logging. We present FoodLogAthl-218, a benchmark constructed from real FoodLog Athl application records and designed to evaluate food understanding under these conditions. The benchmark contains 6,925 images, 14,349 dish bounding boxes, and 218 consolidated classes derived from 11,460 meal logs collected over time, with anonymized user and timestamp metadata that support temporally ordered evaluation. Its construction pipeline combines automated filtering, manual review, and post-hoc label consolidation to preserve real logging conditions while removing invalid or privacy-sensitive content. We define three tasks: closed-set dish classification from crops, continual user-aware adaptation along the time axis, and menu-aware classification for dishes appearing in multi-dish meals. Across representative CLIP-style and large multimodal model baselines, the benchmark remains challenging: GPT-4o reaches 0.4243 Recall@1 and 0.3732 F1, while CLIP ViT-L/14 attains the best Recall@5 of 0.6000. Feature-space analyses against UEC-Food 256 show higher intra-class spread and lower cluster separability, consistent with greater realism and difficulty. Chronological adaptation substantially improves performance over zero-shot inference, and menu-level descriptions increase F1 and Precision@1 while reducing recall, revealing an informative precision-recall trade-off. These results position FoodLogAthl-218 as a benchmark for evaluating robustness, personalization, and context use in real-world food recognition.

1 Introduction

Food image understanding supports dietary assessment, food logging, retrieval, and recipe understanding, and has motivated a long line of benchmark construction efforts in food computing [2, 9, 17, 18, 21]. However, the data distribution used to evaluate recognition systems is still often substantially easier than the one faced by deployed dietary logging applications. Images in widely used datasets are typically collected from the web or other curated sources, contain a single salient dish, and are organized around a predefined class list [2, 4, 6, 18, 26]. This design is useful for standardization, but it underrepresents several properties that matter in practice: severe class imbalance, casual phone photography, co-occurring dishes, evolving user-specific histories, and labels that originate from a logging workflow rather than from a benchmark ontology.

FoodLogAthl-218 addresses this evaluation gap by starting from real meal logs and consolidating them into a public benchmark after collection. The resulting benchmark preserves the bottom-up structure of dietary records: meal photos come from ordinary users, many images contain multiple dishes, and the benchmark retains temporal continuity through anonymized user IDs and timestamps. This allows evaluation settings that are difficult to realize with traditional static datasets, including chronological adaptation and menu-aware dish recognition.

The main contribution of this paper is not a new recognition model, but a benchmark and protocol that expose different failure modes of current systems. The benchmark is built from 11,460 app records collected between May 2023 and October 2024 and curated into 6,925 images, 14,349 dish bounding boxes, and 218 consolidated classes. We validate that the data are both realistic and challenging: the class histogram is strongly long-tailed, the images reflect informal private-use photography, and feature-space analyses show weaker class separation than in UEC-Food 256 [9]. We then benchmark representative CLIP-style models [19, 23], open LMMs [1, 13, 14], and GPT-4o-style closed models using a common evaluation setup implemented with LMMs-Eval [11, 28].

Empirically, FoodLogAthl-218 is difficult for both retrieval-style and generative vision-language systems. In the closed-set task, the strongest baseline reaches only 0.4243 Recall@1, while temporal adaptation improves accuracy substantially over zero-shot inference, and menu-aware descriptions shift the precision-recall balance rather than uniformly improving every metric. These outcomes illustrate the kind of diagnostic signal the benchmark is intended to provide: not which model "wins" in isolation, but how different systems behave under realistic logging conditions.

Our contributions are threefold. First, we introduce FoodLogAthl-218, a benchmark built from dietary management application logs with image crops, timestamps, anonymized users, and multi-dish meal context. Second, we document a mixed automated-human curation pipeline, including filtering, outlier detection, manual review, and label consolidation. Third, we provide three evaluation tasks and baseline results that reveal the difficulty of real-world food recognition, the benefit of temporal adaptation, and the trade-offs introduced by contextual descriptions.

2 Related Work

2.1 Food Recognition Benchmarks

Food recognition has been studied for more than a decade, from early datasets such as PFID [3] and UEC-Food [4, 9] to larger web-scale resources such as Food-101 [2], UPMC Food-101 [26], VegFru [6], FoodAI-756 [21], and Food2K [18]. These resources have been valuable for measuring fine-grained recognition performance, but they are usually label-first datasets: classes are specified in advance and images are then collected to populate them. This tends to favor balanced or curated distributions and discourages retaining noisy or ambiguous multi-dish compositions.

Recent food-domain benchmark work has also expanded beyond classification. Food-500 Cap studies caption-based probing for food vision-language models [16], while datasets such as Recipe1M and later cross-modal resources emphasize recipe retrieval or generation [22, 24, 26]. FoodLogAthl-218 complements these efforts

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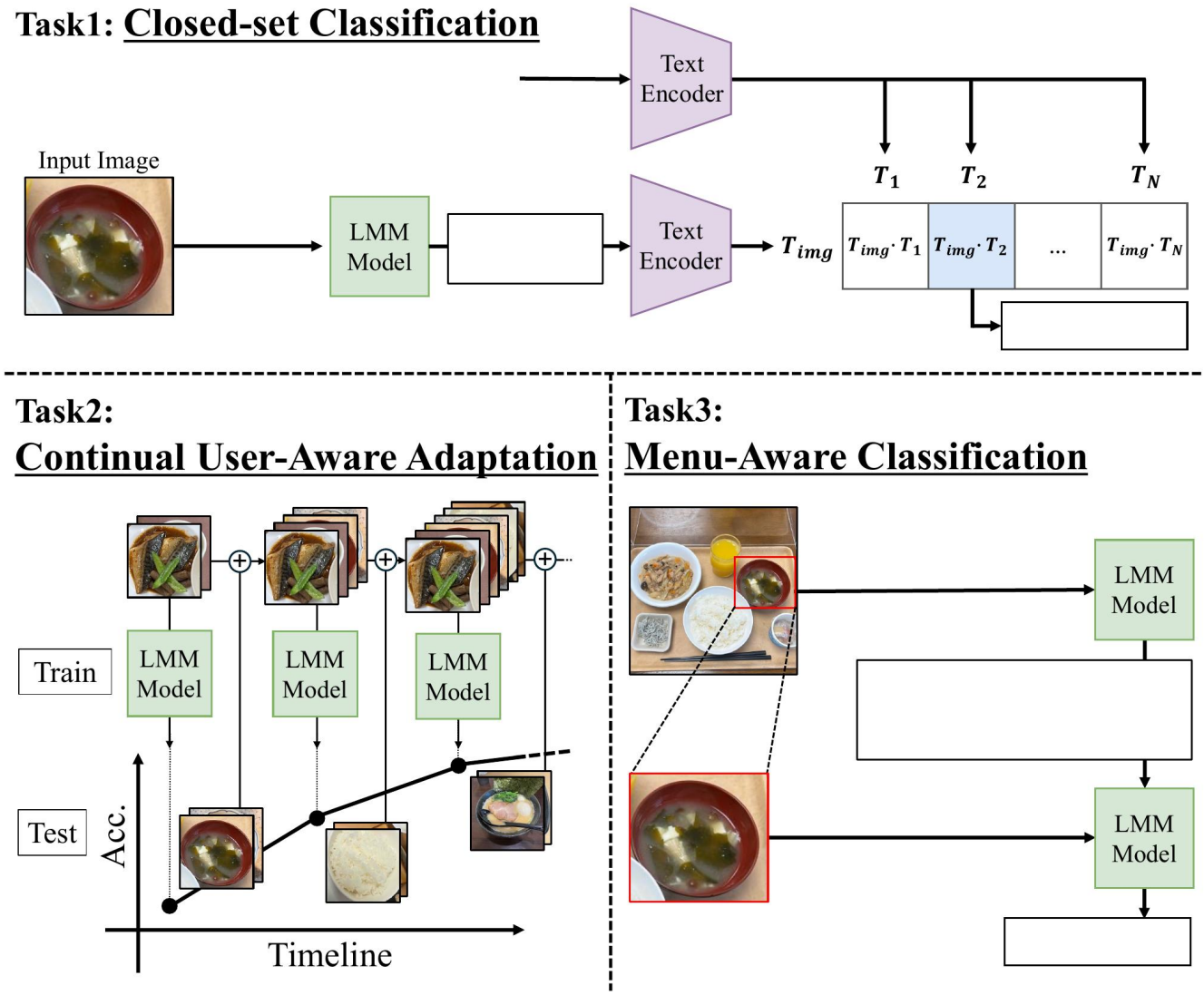

Figure 1: Overview of the three tasks supported by FoodLogAthl-218. Task 1 evaluates closed-set dish classification on cropped dish regions. Task 2 evaluates continual user-aware adaptation on temporally ordered data. Task 3 evaluates menu-aware classification, where whole-meal context is used to help classify a cropped dish from a multi-dish image.

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by focusing on evaluation conditions arising in dietary logging: temporal continuity, strong imbalance, and meal-level context.

2.2 Real-World Logging and Personalization

Several prior works have argued that personalized food recognition differs from closed, static evaluation. Earlier FoodLog Athl studies showed that user-specific history and incremental adaptation can improve recognition in realistic logging scenarios [5, 10, 27]. Our benchmark extends this line of work from a personalization setting to a public evaluation protocol: instead of proposing a new personalization algorithm, we provide a benchmark with temporally ordered data and representative adaptation baselines so future work can measure these effects reproducibly.

2.3 Vision-Language Models for Food Understanding

Large vision-language models have expanded the design space for food understanding, including captioning, recipe generation, and open-vocabulary recognition [8, 12–14, 16, 19]. At the same time, broad web pretraining may obscure weaknesses on domain-specific, real-world imagery. Food-500 Cap explicitly showed that food-specific evaluation can reveal limitations not visible on general-purpose benchmarks [16]. FoodLogAthl-218 serves a similar purpose, but emphasizes user-submitted meal logs and temporal evaluation rather than caption probing alone.

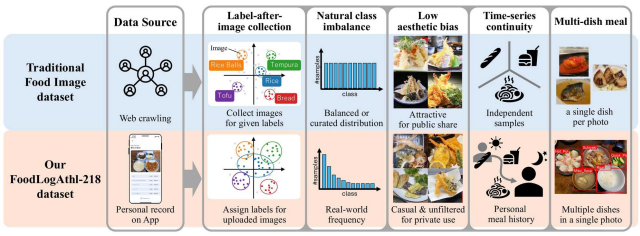

Figure 2: Conceptual comparison between conventional food datasets and FoodLogAthl-218. The benchmark is constructed from personal logs, then labeled and consolidated afterward, which preserves natural imbalance, casual imagery, temporal continuity, and multi-dish meal structure.

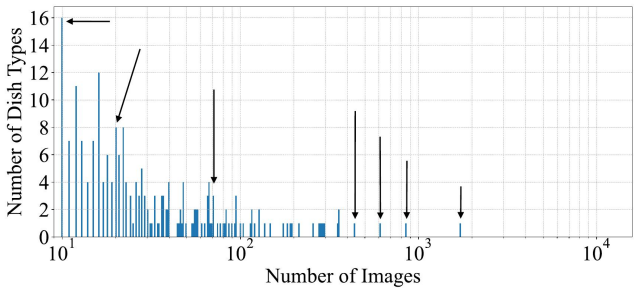

Figure 3: Class-frequency histogram of FoodLogAthl-218. The distribution is strongly long-tailed, with only a small number of common classes and many low-frequency dish types.

3 Benchmark Design and Validation

3.1 Data Source and Curation Pipeline

FoodLogAthl-218 is constructed from records submitted through the FoodLog Athl dietary management application. The raw collection contains 11,460 images, 61,414 bounding boxes, and 5,889 raw labels from 632 users between May 2023 and October 2024. Each record includes an image, anonymized user ID, date and time, dish names linked to an approximately 140K-item food database, and system-provided detections that users could correct during logging.

The benchmark is built bottom-up from these records, treating each image as an atomic meal record so that multi-dish composition is preserved. Table 1 summarizes the filtering process. First, records without valid images or boxes were removed, reducing the pool to 11,285 images and 26,065 boxes. Second, classes with fewer than 10 samples were discarded, yielding 9,121 images across 1,180 classes. Third, CLIP ViT-B/16 features [19] were used for classwise outlier filtering: samples more than 1.5 standard deviations from the class centroid were flagged, after which manual review corrected or removed invalid boxes and labels. The curation policy removed whole images rather than single instances when necessary, so meal context remained intact. Finally, GPT-4o-assisted grouping followed by human adjudication consolidated synonyms and near-duplicates into 218 final classes.

After curation, the released benchmark contains 6,925 images, 14,349 bounding boxes, 218 classes, and 513 users. According to the project documentation, release preparation also included a two-stage manual privacy review, informed consent, and ethics oversight. We do not report inter-annotator agreement because the available artifacts do not contain it; this remains a limitation of the current release.

3.2 What the Benchmark Measures

FoodLogAthl-218 is designed to measure recognition performance under three properties that are common in dietary logging but weakly represented in many standard datasets. First, labels are assigned after image collection, rather than by collecting images for a fixed ontology. Second, class frequencies remain highly imbalanced, reflecting what users actually log. Third, the benchmark preserves both temporal continuity and multi-dish context. Figure 2 contrasts this setup with more traditional food datasets.

The class histogram in Figure 3 shows a pronounced long tail. Many classes appear only a small number of times, while a few everyday foods dominate the sample count. This is precisely the regime in which retrieval-style models and general-purpose LMMs are often deployed in practice but are rarely benchmarked in a realistic manner.

3.3 Validation of Realism and Difficulty

We validate the benchmark in two ways. First, the qualitative artifacts show that images include casual phone captures, private-use meals, convenience-store items, and multiple dishes in a single frame, matching the target use case rather than an aesthetically curated photography distribution. Second, the quantitative feature-space analysis in Table 2 and the provided diversity metrics indicate weaker cluster structure than in UEC-Food 256 [9]. Under CLIP features, FoodLogAthl-218 has higher trace (81.70 vs. 71.21), lower silhouette (-0.0210 vs. 0.0335) [20], higher Davies–Bouldin (4.52 vs. 3.64), and lower Calinski–Harabasz (27.51 vs. 74.05), with the same qualitative pattern under EVA-CLIP and Swin [15, 23]. These statistics support the intended interpretation: the benchmark is not only long-tailed, but also visually less separable than a standard web-style benchmark.

4 Tasks and Experimental Protocol

FoodLogAthl-218 supports the three tasks illustrated in Figure 1.

Task 1: Closed-set dish classification. Each cropped dish region is assigned one of 218 benchmark labels. Performance is reported using F1 and Recall@1/3/5. The label space is fixed, but the visual evidence remains difficult because crops often contain partial dishes, clutter, or visually similar classes.

Task 2: Continual user-aware adaptation. Samples are ordered chronologically by logging time. Models are updated over time and evaluated week by week, allowing the benchmark to measure whether accumulating records improve future recognition. The provided plots report weekly accuracy together with a weighted one-month moving average over a ±2 week window, where weights are proportional to the weekly sample count.

Task 3: Menu-aware classification. This task asks whether whole-meal context helps classify a cropped dish in a multi-dish photo. The model first produces a free-form description of the

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Table 1: Data Filtering Process and the Number of Images, Bounding Boxes (bboxes) and Classes.

Step	Operation	#images	(diff)	#bboxes	(diff)	#classes	(diff)
0	Raw data	11,460		61,414		5,889	
1	Remove samples w/o image	11,460	(0)	26,646	(-34,768)	3,167	(-2,722)
2	Remove samples w/o bounding box	11,285	(-175)	26,065	(-581)	3,071	(-96)
3	Remove classes w/ <10 samples	9,121	(-2,164)	21,944	(-4,121)	1,180	(-1,891)
4	Remove outliers based on image features	7,537	(-1,584)	17,513	(-4,431)	1,180	(0)
5	Manual review and remove invalid images	6,925	(-612)	14,349	(-3,164)	1,180	(0)
6	Merge similar classes and clean labels	6,925	(0)	14,349	(0)	218	(-982)

Table 2: Classification performance on FoodLogAthl-218.

Model	F1	R@1	R@3	R@5
GPT-4o	**0.3732**	**0.4243**	**0.4991**	0.5221
Qwen2.5-VL-7B-Instruct	0.2349	0.3684	0.4609	0.4859
Qwen2-VL-7B-Instruct	0.1594	0.3656	0.4590	0.4781
Qwen2-VL-3B-Instruct	0.1386	0.2663	0.3250	0.3517
llava-onevision-qwen2-7b-ov	0.1546	0.3629	0.4497	0.4639
EVA-CLIP (ViT-g)	0.2465	0.2654	0.4897	0.5944
CLIP (ViT-L/14)	0.2393	0.2567	0.4885	**0.6000**
llava-v1.6-34b	0.1354	0.2432	0.3779	0.4063
llava-v1.5-7b	0.0991	0.2140	0.2721	0.2973
llava-v1.6-mistral-7b	0.0944	0.1932	0.2595	0.2831

full image, then classifies the crop using both the crop and the generated description. Because model outputs are open-ended, the final class prediction is obtained by embedding both the generated dish description and the candidate class names with E5 [25] and choosing the label with highest cosine similarity.

The baseline suite spans CLIP and EVA-CLIP retrieval models [19, 23], LLaVA-family models [13, 14], Qwen-family models [1], and GPT-4o-style proprietary multimodal inference, with evaluation run through LMMs-Eval [11, 28]. For the adaptation experiments, the provided figures compare zero-shot inference, full fine-tuning, and LoRA-based tuning [7]. We report the exact tabled values where available and describe plot-based trends qualitatively when the released artifacts do not provide raw numeric logs.

5 Benchmark Results and Analysis

5.1 Task 1: Closed-Set Classification

Table 2 reports the main closed-set results. FoodLogAthl-218 is challenging for all tested systems. GPT-4o achieves the best F1 (0.3732) and the best Recall@1 (0.4243), while CLIP ViT-L/14 yields the strongest Recall@5 (0.6000). EVA-CLIP shows a similar pattern, with relatively modest Recall@1 but competitive higher-rank recall. This suggests that contrastive models often retrieve plausible alternatives but struggle to rank the correct fine-grained dish first.

Open LMMs trail GPT-4o by a substantial margin. Qwen2.5-VL-7B-Instruct is the strongest open model in the table, reaching 0.2349 F1 and 0.3684 Recall@1, but still falls short of the proprietary baseline. More broadly, the gap between Recall@1 and Recall@5 for most models indicates that the benchmark penalizes subtle confusions among visually similar foods rather than only catastrophic failures. This is consistent with the higher feature-space overlap discussed in Section 3.

This ranking pattern is important for interpreting benchmark difficulty. In FoodLogAthl-218, some systems appear able to retrieve semantically nearby dishes but fail to place the correct class first, which separates candidate-retrieval ability from top-rank calibration. That distinction matters for deployment: an assisted logging interface may still be useful if the right dish is reliably surfaced in a short list, whereas automatic logging requires much stronger first-choice accuracy. Reporting both rank-based recall and F1 therefore makes the benchmark informative for distinct operating regimes instead of rewarding only one style of system behavior.

5.2 Task 2: Continual User-Aware Adaptation

The chronological evaluation exposes a different property of the benchmark: performance is not stationary, and access to prior user data matters. Figure 4 compares zero-shot inference with full fine-tuning and LoRA adaptation on all users. The zero-shot moving average starts near 0.23–0.24 and drifts downward to roughly 0.15–0.18 late in the stream, while both adaptation methods improve steadily. Full fine-tuning rises from about 0.24 to roughly 0.49 by the end of the timeline, and LoRA tracks a similar trajectory, improving from about 0.26 to roughly 0.48. The main conclusion is qualitative but clear: temporally accumulating logs produce a gain of more than 30 percentage points over zero-shot inference.

Figure 4 also shows that user-specific adaptation can reveal behavior hidden by global averages. For a heavy logger (User A), LoRA tuned only on that user's data overtakes the all-user LoRA model for much of the middle portion of the timeline, reaching moving-average accuracy above 0.6, before the curves narrow and partially reverse near the end. This indicates that personalization is beneficial, but that the relative value of personal versus pooled data can change over time as more shared data become available.

5.3 Task 3: Menu-Aware Classification

Table 3 evaluates whether whole-meal descriptions improve crop classification. Adding menu-level description raises F1 from 0.1594 to 0.2304 and Precision@1 from 0.1019 to 0.1788, but lowers Recall@1 from 0.3656 to 0.3240 and similarly reduces Recall@3 and Recall@5. The benchmark therefore reveals a meaningful trade-off: context makes top predictions more precise, but it can also overcommit the model to a narrower set of hypotheses.

The released examples help explain this behavior. In some cases, the whole-image description introduces useful dish relations and resolves ambiguity in the crop. In other cases, the description injects

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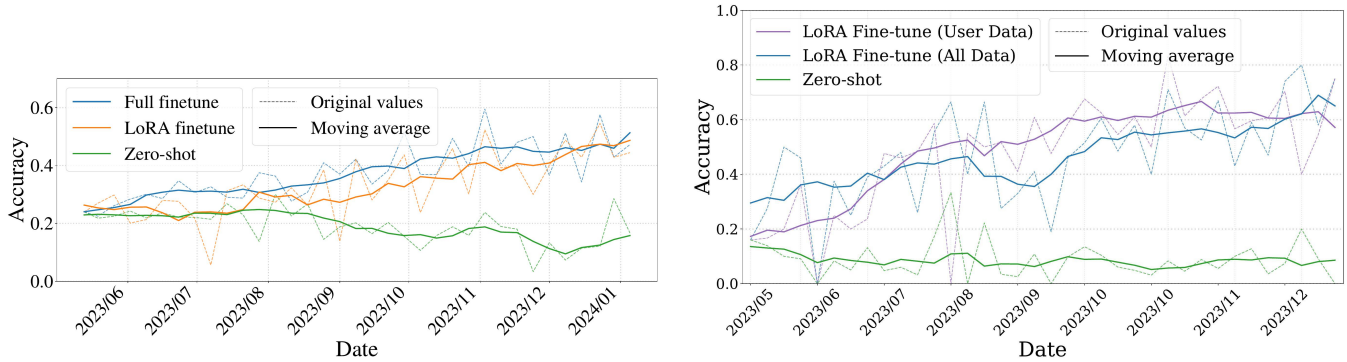

Figure 4: Continual adaptation results. Left: weekly accuracy and weighted moving averages for zero-shot inference, full fine-tuning, and LoRA fine-tuning on all users. Right: zero-shot, all-user LoRA, and user-specific LoRA for User A. Both plots show strong gains from chronological adaptation, and the user-specific plot indicates that personalization can outperform pooled adaptation for extended periods of the stream.

Table 3: F1 score, Precision@1 (P@1), and Recall@K (R@1, R@3, R@5) for Qwen2.5-VL-7B-Instruct on the description-guided classification task.

	F1	P@1	R@1	R@3	R@5
w/o description	0.1594	0.1019	**0.3656**	**0.4590**	**0.4781**
w/ description	**0.2304**	**0.1788**	0.3240	0.4032	0.4336

a plausible but wrong meal narrative that steers the model away from the correct class. FoodLogAthl-218 is therefore diagnostic not only for whether models use context, but for *how* they use it.

Taken together, the three tasks show that FoodLogAthl-218 is diagnostic along multiple axes rather than reducible to a single leaderboard number. Task 1 measures fine-grained visual discrimination under a fixed ontology, Task 2 measures whether models can benefit from temporally accumulated user history, and Task 3 measures whether contextual meal evidence is used helpfully or overconfidently. The fact that the benchmark exposes different patterns across these settings is itself an important result: a model with strong closed-set retrieval is not necessarily the one that benefits most from adaptation or menu context. This cross-task instability is precisely the kind of evaluation signal that is easy to miss in static, single-setting food benchmarks.

6 Discussion and Conclusion

FoodLogAthl-218 is intended as a benchmark for evaluating food recognition under realistic dietary logging conditions rather than as a comprehensive measure of all food understanding. Its main strengths are the use of real app data, the preservation of temporal and meal-level context, and the combination of three tasks that probe complementary capabilities: fine-grained recognition, adaptation over time, and context use. The benchmark shows that strong general-purpose multimodal systems still struggle on casual food logs, that feature-space separability is weaker than in standard food datasets, and that both personalization and context can help, though not uniformly across metrics.

The benchmark is also useful for studying distribution shift within a single application setting. Breakfast staples, packaged lunches, and restaurant meals can recur with different frequencies and co-occurrence patterns over time, so models may fail even when the class vocabulary is unchanged.

These findings also suggest a concrete reporting recommendation for future use of the benchmark. Because Task 1, Task 2, and Task 3 reward different behaviors, a single aggregate score would hide whether a system is strong because it memorizes common classes, adapts effectively to repeated users, or uses meal context conservatively. We therefore view FoodLogAthl-218 as a slice-aware evaluation suite: papers should report rank-based retrieval metrics and F1 for Task 1, time-resolved curves for Task 2, and precision-recall changes for Task 3, rather than collapsing performance into one number. This protocol makes improvements easier to interpret and helps separate genuine robustness gains from gains that help only one operating regime.

The benchmark also has clear limitations. The class set is restricted to labels that remain after filtering and consolidation, so rare foods are underrepresented in the final 218-way task even though they appear in the raw logs. Inter-annotator agreement is not available in the released artifacts. Label merging improves usability but may smooth over distinctions that matter for some nutritional or culinary applications. Finally, although the benchmark is more contamination-resistant than a web-sourced dataset, contamination from general web pretraining cannot be ruled out for large foundation models.

Within these limits, FoodLogAthl-218 provides a useful measurement instrument. It enables controlled comparison on a realistic distribution, exposes large gaps between zero-shot and adapted

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performance, and highlights that menu context changes model behavior in nuanced ways rather than simply improving every metric. We hope the benchmark supports future work on robust, personalized, and context-aware food recognition systems for everyday dietary logging.

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